

RADHEY KALRA

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Open source contributor and full-stack developer working across ML/audio systems, Linux tooling, and self-hosted infrastructure.

EDUCATION

Bachelor of Technology in CSE

Model Institute of Engineering and Technology

Aug. 2023 – May 2027

Jammu and Kashmir, India

Schooling (Class I – XII)

Air Force School, Jammu

2011 – 2023

Jammu and Kashmir, India

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, Go, Rust, SQL, Bash, PowerShell, QML, Vala

Frameworks & Platforms: SvelteKit, React, Astro, Tailwind CSS, React Native, Bun

Developer Tools: Git, Docker, CI/CD, Cloudflare, Vercel, Kubernetes, Terraform

Databases & Cloud: Convex, PostgreSQL, MySQL, AWS, GCP

Domains: Audio Processing, Linux Servers, Full-Stack Development

PROJECTS

BoxBox

Nov. 2025 – Present

Self-hosted file management system for Linux servers | [Website](#) | [GitHub](#) | [Docker](#)

- Built a lightweight self-hosted file manager in **Go + Svelte** for multi-mount browsing, chunked/resumable uploads, live job progress, media preview/streaming, and in-browser editing via **Monaco**.
- Ships as a **31MB single binary** with the UI embedded, plus a published Docker image (**latest**) for one-command homelab/NAS deploys.
- Grew the project to **200+ GitHub stars** and **5K+ downloads**.

U-Net Audio Filter

Jul. 2025 – Present

Audio enhancement by removing background noise | [GitHub](#)

- Building an audio enhancement system with a **custom U-Net-inspired architecture** and end-to-end audio denoising pipeline.
- Implemented **Python/PyTorch + audio DSP** preprocessing, spectrogram modeling, and evaluation tooling, with early results showing minor gains over open-source baselines.

HPomen Linux (Victus Control)

Mar. 2026 – Jul. 2026

Reverse-engineering HP OMEN control utility behavior on Linux | [GitHub](#) | [AUR](#)

- Built a Linux-first OMEN/Victus control suite in **Vala** with a GTK4 monitor app, tray companion, D-Bus helper, and probe CLI.
- Reverse-engineered **HP WMI/sysfs** behavior to validate profile/fan controls (**auto/max**) and extend **hp-wmi** with capability-gated safety fallbacks.
- Submitted a Linux kernel **hp-wmi** patch accepted into **platform-driver-x86**, adding Victus 15-fb0xxx board support for fan-control feature data; also provided **Tested-by** validation for another **hp-wmi** fan-control patch on real hardware.

Fast Blob Storage (FBS)

Apr. 2026 – May 2026

Self-hosted S3-compatible object storage for single-node / homelab deploys | [Live](#) | [GitHub](#) | [Docker](#)

- Built a single-node **S3-compatible** object store in **Go** (filesystem blobs + **SQLite** metadata) with a **SvelteKit** dashboard, shipped as a single binary and Docker image (**latest**).
- Features dual auth (**SigV4** + bearer), multipart uploads, a management API, and zero-copy Linux **sendfile()** egress for fast local downloads without distributed-storage overhead.
- Benchmarks: **~15MB** binary, **~14–16MB** idle RSS, up to **~1.3 GB/s** downloads / **~3.8 GB/s** concurrent aggregate—about **12–14×** less memory than MinIO; **50+ GitHub stars**.

Additional Projects

- **QckPages** — SvelteKit/Convex app for publishing HTML pages, FBS-backed storage ([Live](#) | [GitHub](#))
- **rsclip** — Rust Wayland clipboard manager with a low-memory daemon, GTK4 UI, SQLite history, search, filters, OCR Support and Secrets management ([GitHub](#) | [AUR](#))
- **Dots-Niri** — Replicates my NiriWM rice on Arch-based install with a single **curl** command ([Website](#) | [GitHub](#))
- **HowTo** — Astro tutorials site for practical hardware and software walkthroughs ([Live](#) | [GitHub](#))

- **Pico32** — C++/PlatformIO serial TUI for ESP32 Wi-Fi and Bluetooth pentesting workflows ([Website](#) | [GitHub](#))
- **TL;DR** — Electron app that summarizes WhatsApp chats with OpenRouter compatible AI models ([GitHub](#))
- **Hotel Booking** — Svelte/Express/MySQL booking platform with authentication and admin workflows ([GitHub](#))
- **Tenant Manager** — React Native app for managing tenants and tracking electricity usage ([GitHub](#))

LEADERSHIP, CO-CURRICULAR ACTIVITIES

Cloud Lead

Aug. 2025 – Present

GDGoC (Google Developer Groups on Campus), MIET Jammu

Jammu and Kashmir, India

- Led cloud learning initiatives through hands-on **Google Cloud** workshops and guided sessions for **100+ students**, and built a tutorial platform ([Live/GitHub](#)) with **20+** lab walkthroughs, notes, and video resources to support **Google Cloud Study Jam** completion.

Cloud and DevOps Lead

Jan. 2026 – Present

AWS Cloud Club MIET Jammu

Jammu and Kashmir, India

- Lead cloud and DevOps initiatives through hands-on workshops, technical sessions, and collaborative projects, driving adoption of **AWS**, containerization, and infrastructure best practices focused on deployment automation and scalability.

IDE Bootcamp Edition 3 Team Leader – 2nd Place

Apr. 2026

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

GNDEC, Ludhiana, Punjab

- Led a team to secure **2nd Place** at the AICTE/MoE Innovation Cell bootcamp by applying rapid prototyping and business modeling to develop an AI-powered B2B SaaS product for automated presentation generation from user-uploaded templates and documents.

IDE Bootcamp Edition 2 Team Leader

Feb. 2025

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

IIT Jammu

- Participated in a 5-day immersive bootcamp at IIT Jammu with Wadhvani Foundation, AICTE, and MoE's Innovation Cell, gaining business modeling, product design, and entrepreneurship experience while leading a team to develop a "Sports Match Finder" app using design thinking and the Lean Canvas model.

Hackathon Team Leader

Internal Hackathon & Smart India Hackathon (SIH 2024 & 2025)

MIET, Jammu

- Led multidisciplinary teams in MIET's internal hackathon and Smart India Hackathon selection rounds in **2024** and **2025**, reaching the **top 9 teams in the college** both years through collaborative problem-solving, rapid prototyping, and real-world solution design.

CERTIFICATIONS & PROGRAMS

Google Cybersecurity - Coursera | [Certificate](#)

Cisco Cybersecurity Operations Fundamentals - Coursera | [Certificate](#)

Palo Alto Networks Cybersecurity - Coursera | [Certificate](#)

Nptel Ethical Hacking - IIT Kharagpur | [Certificate](#)

Nptel Deep Learning - IIT Ropar | [Certificate](#)

IBM SkillsBuild: AI Agent Architect | [Certificates](#)

IDE Bootcamp (AICTE / MoE) | [Certificate](#)